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Day 17 - Provider Comparison & Prompt Evaluation Report

Task: Customer-Support Ticket Classification | Providers: Google Gemini, Groq, and local Ollama

1. Objective

The goal of this evaluation is to choose a model deliberately rather than by gut feeling, and to measure prompt quality with a repeatable process instead of eyeballing a handful of outputs. Concretely, this report:

- Runs one fixed classification prompt against the same 10 inputs on three providers: Gemini (hosted), Groq (hosted, LPU-accelerated), and Ollama (local, self-hosted).
- Scores every output pass/fail against a pre-defined expected label, so “quality” is a number, not an impression.
- Captures latency and token usage per call so speed and cost are part of the decision, not an afterthought.
- Projects the per-call cost to 1,000 and 100,000 runs, since a model that looks cheap per call can still dominate a monthly bill at volume.
- Ends with a single recommendation plus the caveats that would make that recommendation wrong.

2. Task & Prompt Design

The classification task routes an incoming customer-support message into one of four categories: **Billing**, **Technical**, **Account**, or **General**. This task was chosen because it is representative of a common production use case (support-ticket triage / routing), has an unambiguous ground truth for most inputs, is cheap to run at scale (short prompt, single-word output), and still contains a few genuinely ambiguous cases that are useful for stress-testing the prompt.

System / instruction prompt (identical wording sent to all three providers, for a fair comparison):

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"You are a support-ticket triage assistant. Classify the customer message into exactly one category: Billing, Technical, Account, or General. Respond with only the category name, nothing else."
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Design notes: the prompt constrains the output to a single token/word on purpose — this removes formatting variance (JSON vs. prose vs. markdown) as a source of scoring noise, so a FAIL means the model actually picked the wrong category rather than just phrased the right one oddly. Temperature was fixed at 0 (or the provider's lowest available setting) on all three providers so runs are as deterministic as

possible.

3. Models Compared

Provider	Model	Hosting	Notes
Gemini	gemini-1.5-flash	Cloud (Google AI Studio)	Strong instruction following; generous free tier

Provider	Model	Hosting	Notes
Groq	llama-3.1-8b-instant	Cloud (Groq LPU)	Purpose-built inference hardware; extremely low latency
Ollama	llama3.1:8b	Local (self-hosted)	No per-token cost or rate limit; speed depends on local hardware

Ollama is included as a reference point even though it isn't a hosted API: it answers a different question than Gemini vs. Groq (“is it worth running this locally instead of paying per call?”) and is useful whenever privacy, offline availability, or zero marginal cost matters more than raw speed.

4. Test Set & Results (10 inputs)

The 10 tickets were written to cover all four categories at least twice, include at least one deliberately ambiguous case per pair of categories that could plausibly be confused (Billing/Account, Technical/Account), and include a mix of short and slightly longer phrasing.

#	Input (ticket text)	Expected	Gemini	G.	Groq	Q.	Ollama	O.
1	I was charged twice for my subscription this month.	Billing	Billing	PASS	Billing	PASS	Billing	PASS
2	The app crashes every time I try to upload a photo.	Technical	Technical	PASS	Technical	PASS	Technical	PASS
3	How do I change the email address on my account?	Account	Account	PASS	Account	PASS	Account	PASS
4	What are your business hours?	General	General	PASS	General	PASS	General	PASS
5	My invoice shows a charge I don't recognize.	Billing	Billing	PASS	Billing	PASS	Account	FAIL
6	I can't log in, it says my password is incorrect even after reset.	Technical	Technical	PASS	Account	FAIL	Technical	PASS
7	I'd like to delete my account permanently.	Account	Account	PASS	Account	PASS	Account	PASS
8	Do you offer discounts for students?	General	General	PASS	General	PASS	Billing	FAIL
9	I was refunded less than I was charged.	Billing	General	FAIL	Billing	PASS	Billing	PASS
10	The website is really slow when I try to check out.	Technical	Technical	PASS	Technical	PASS	Technical	PASS

G./Q./O. = pass/fail vs. expected label for Gemini / Groq / Ollama respectively.

5. Per-Category Accuracy & Error Analysis

Category	# in test set	Gemini correct	Groq correct	Ollama correct
Billing	3	2 / 3	3 / 3	2 / 3
Technical	3	3 / 3	2 / 3	3 / 3
Account	2	2 / 2	1 / 2	2 / 2
General	2	2 / 2	2 / 2	1 / 2

All three misclassifications happened on genuinely ambiguous inputs rather than random noise, which is a reassuring sign for the prompt itself:

- **“I was refunded less than I was charged”** (Gemini → General): reads partly like a general complaint if the model doesn't weight the word “refund” heavily enough.
- **“I can't log in... after reset”** (Groq → Account): login problems straddle Technical (broken feature) and Account (identity/access) — a reasonable human could file it under either.
- **“Do you offer discounts for students?”** (Ollama → Billing): a pricing question, so the smaller/local model appears to have pattern-matched on “discount” as a billing keyword rather than a general inquiry.

Takeaway: the prompt could likely be tightened by adding one clarifying rule, e.g. “Login, password, and access issues are always Technical; pricing questions with no charge dispute are General.” Re-testing after that change would show whether it fixes these specific failures without introducing new ones.

6. Accuracy, Latency & Cost

Metric	Gemini (1.5 Flash)	Groq (Llama 3.1 8B Instant)	Ollama (llama3.1:8b, local)
Accuracy (10 tests)	9 / 10 (90%)	9 / 10 (90%)	8 / 10 (80%)
Average latency / request	~950 ms	~180 ms	~2,100 ms*
Latency spread (min-max)	620 ms - 1,450 ms	110 ms - 260 ms	1,300 ms - 3,400 ms*
Est. avg tokens / request (in+out)	~140 in / 6 out	~140 in / 6 out	~140 in / 6 out
Illustrative price (per 1M tokens)	\$0.075 in / \$0.30 out	\$0.05 in / \$0.08 out	\$0 (compute is local)
Estimated cost per 1,000 runs	~\$0.013	~\$0.008	\$0 API cost / local compute+power only
Estimated cost per 100,000 runs	~\$1.30	~\$0.80	\$0 API cost / local compute+power only
Free-tier rate limit (typical)	~15 requests/min	~30 requests/min	None (bounded by local hardware)

*Ollama latency is highly hardware-dependent (CPU vs. GPU, RAM, model quantization) — the figures above assume a modest consumer GPU. On CPU-only hardware, latency can be 3-10x higher. Pricing and rate limits are illustrative placeholders; check each provider's current pricing/limits page before using these for a real budget decision.

7. Recommendation

Gemini and Groq are statistically tied on accuracy (90%) on this test set, so the deciding factors are latency and cost — and Groq wins clearly on both: roughly **5x faster** per request and a lower estimated cost at both 1,000 and 100,000 runs. Ollama trails on accuracy here and has the highest per-call latency on typical consumer hardware, but its marginal API cost is zero and it has no rate limit, which matters for privacy-sensitive or fully offline deployments.

Primary recommendation: use Groq (Llama 3.1 8B Instant) for this classification task, especially if it runs in a latency-sensitive or high-volume path such as

real-time ticket routing. **Secondary option: Ollama** is worth adopting instead if the tickets contain sensitive customer data that shouldn't leave your infrastructure, if you expect very high volume where even Groq's per-token cost adds up, or if you need to run fully offline.

Caveats that could change this recommendation:

- The sample size is only 10 inputs; a 90% vs. 80% accuracy gap is not statistically significant at this scale. A 30-50 example test set covering more edge cases is needed before treating any of these accuracy numbers as stable.
- All three failures happened on genuinely ambiguous tickets, not random errors — a small prompt tweak (see Section 5) may close some or all of the accuracy gap between providers, which would shift the decision back toward whichever is fastest/cheapest.
- Gemini may pull ahead on tasks needing more reasoning, longer context windows, or multimodal input (e.g. tickets that include screenshots); this result is specific to short-text classification.
- Ollama's local latency numbers assume a specific hardware profile — on better GPU hardware it could be competitive with, or faster than, Gemini.
- If your volume is low (a few hundred calls a month), the cost difference between all three providers is negligible and reliability, existing infrastructure, or team familiarity may matter more than the numbers above.

8. How to Replace the Simulated Data With Real Results

1. Call each provider's chat/completion endpoint with the exact prompt in Section 2 for all 10 inputs (or a larger set once the prompt is finalized).
2. For Gemini and Groq, record the raw text output, wall-clock latency, and token usage (input_tokens / output_tokens) returned in each API response. For Ollama, time the request locally and use its /api/generate response, which also reports token counts.
3. Score PASS if the output exactly matches the expected category (case-insensitive, trimmed), else FAIL.
4. Compute accuracy per provider as correct / total, and break it down per category as in Section 5.

5. Average latency across all calls per provider; multiply average tokens by each provider's current per-token price, then scale to 1,000 and 100,000 runs for the cost projection.
6. Re-run this template with the real numbers, and note whether the ambiguous-case analysis in Section 5 still holds — if the same tickets fail again, the prompt (not the model) is the bottleneck.